

Ellipsoids and their Volume

$$A = \begin{pmatrix} r_1 & & \\ & r_2 & \\ & & \ddots \\ & & & r_d \end{pmatrix} \quad \begin{aligned} x^T A^{-1} x &\leq 1 \\ \frac{1}{r_1} x_1^2 + \dots + \frac{1}{r_d} x_d^2 &\leq 1 \\ \frac{1}{4} x_1^2 + \frac{1}{9} x_2^2 &\leq 1 \end{aligned}$$


In general: $A \succ 0$, $c \in \mathbb{R}^n$

$$\begin{aligned} E &= E(A, c) \triangleq \{x \in \mathbb{R}^n : (x-c)^T A^{-1} (x-c) \leq 1\} \\ &= A^{1/2} S(0, 1) + c. \end{aligned}$$

$$\text{Vol}(E(A, c)) = \text{Vol}(E(A, 0)) = \sqrt{\det A} \cdot \text{Vol}(S(0, 1))$$

$$\text{Vol}(S(0, 1)) = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)}, \quad \Gamma(x) = \int_0^{\infty} e^{-t} \cdot t^{x-1} dt, \quad x > 0$$

Cutting Ellipsoids

$$E(A, c)$$

$$\mathcal{X} = E(A, c) \cap \{x \in \mathbb{R}^n : a^T x \leq a^T c\}$$

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Thm: The Löwner-John ellipsoid
of $\mathcal{X}$ is: $E(\hat{A}, \hat{c})$

$$\hat{c} = c - \frac{1}{n+1} \cdot z$$

$$\hat{A} = \frac{n^2}{n^2-1} \left(A - \frac{2}{n+1} \cdot z z^T \right)$$

$$z = \frac{1}{\sqrt{a^T A a}} A a$$

Volume Reduction — 1/3

For $E = E(A, c)$ and

$\hat{E} = E(\hat{A}, \hat{c})$, the LJ ellipsoid for $\mathcal{E}$

$$\begin{aligned} \underline{\text{Thm}}: \frac{\text{Vol}(\hat{E})}{\text{Vol}(E)} &= \frac{\sqrt{\det \hat{\lambda}}}{\sqrt{\det \lambda}} \\ &= \left(\left(\frac{n}{n+1} \right)^{n+1} \cdot \left(\frac{n}{n-1} \right)^{n-1} \right)^{1/2} \end{aligned}$$

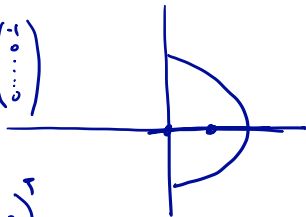
$$\underline{\quad} < e^{-1/2n} \quad < \quad \underline{\quad} 1$$

Volume Reduction — 2/3

Assume: $\underline{E} = \underline{F} = \underline{E}(1, 0)$

Half-space defined by: $\underline{a} = \begin{pmatrix} -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$

$$\underline{z} = \frac{1}{\sqrt{\underline{a}^T \underline{A} \underline{a}}} \underline{A} \underline{a} = \begin{pmatrix} -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$



$$\hat{\underline{c}} = \underline{c} - \frac{1}{n+1} \underline{z} = \left(\frac{1}{n+1}, 0, \dots, 0 \right)^T$$

$$\hat{\underline{A}} = \frac{n^2}{n-1} \left(\underline{I} - \frac{2}{n+1} \underline{z} \underline{z}^T \right) = \text{diag} \left(\frac{n^2}{(n+1)^2}, \frac{n^2}{n^2-1}, \dots, \frac{n^2}{n^2-1} \right)$$

$$\frac{\text{vol}(\hat{\underline{E}})}{\text{vol}(\underline{E})} = \frac{\sqrt{\det(\hat{\underline{A}})}}{\sqrt{\det(\underline{A})}} = \sqrt{\det(\hat{\underline{A}})} = \left(\frac{n^2}{(n+1)^2} \times \left(\frac{n^2}{n^2-1} \right)^{n-1} \right)^{1/2} = \exp. \text{ on prev slide.}$$

can now check: $< \exp(-1/2n) < 1$.

